

Sensor-Based Urban Thermal Modeling and Machine Learning Forecasting in Coburg

Selected data-analysis and machine-learning excerpt from M.Eng. thesis

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Scope note. This document is a selected and condensed summary of the sensor-data, feature-engineering, machine-learning, and forecasting sections of my M.Eng. thesis. The datasets, methods, figures, and numerical results are taken from the original thesis. No new model runs or results are introduced.

Study question and observational design

The thesis examined how urban-rural thermal contrasts vary across Coburg and whether short-term machine-learning models can predict those contrasts from local sensor and reference-station observations. The analysis combined the classical air-temperature difference (ΔT) with a PET-inspired thermal-stress proxy ($\Delta \text{PET}_{\text{proxy}}$), allowing the study to move beyond temperature alone while remaining compatible with the available measurements.



Figure 1. Selected urban stations and 100 m GIS buffers from the original thesis: Marktplatz (M), Albertplatz (A), and Hofgarten (H).

Station	Local setting	Built fraction (100 m)
Marktplatz	Dense historic city center	0.576
Albertplatz	Open urban square / intermediate	0.412
Hofgarten	Urban green space / park	0.002

1. Data basis, harmonization, and thermal indicators

Urban observations came from the City of Coburg sensor network. The selected Senstick SMC30 stations supplied air temperature, relative humidity, and air pressure at a nominal 15-minute cycle. The rural reference was the DWD station at Neuses, providing standardized meteorological forcing including temperature, humidity, wind, pressure, and global radiation. The full analysis window covered January 2024 to August 2025, and the urban and rural datasets were harmonized to a common 15-minute analytical grid.

$$\Delta T = T_{urban} - T_{rural}$$

$$\Delta PET_{proxy} = PET_{proxy, urban} - PET_{proxy, rural}$$

Core contrast variables used throughout the analysis.

PET-inspired thermal-stress proxy

A full Physiological Equivalent Temperature calculation was not possible because detailed urban wind, mean radiant temperature, and complete radiation geometry were not available consistently. A transparent PET-inspired proxy was therefore constructed from the variables that were available. Coefficient specification was performed only on January-June 2024 to keep that step separate from later model development and testing.

$$PET_{proxy}(s) = T_{urban} + 0.10(e_{a, urban} - 10) - 2.0\ln(1 + \alpha_{U, s}U_{rural}) + 0.03(\alpha_{R, s}R_{g, rural}) + 0.30p_{norm}$$

Operational morphology-aware PETproxy formulation used in the thesis. $\alpha_{U,s}$ and $\alpha_{R,s}$ are station-specific wind and radiation adjustment factors.

As a plausibility check, daily mean PETproxy and ERA5-derived UTCI followed a very similar seasonal pattern during January-June 2024, with a Pearson correlation of $r = 0.94$. The UTCI comparison was treated as an external reference rather than as a local ground truth.

Feature engineering

Feature group	Examples / purpose
Atmospheric state	Urban/rural temperature and relative humidity; normalized pressure anomaly
Forcing	Rural wind speed and global shortwave radiation
Cyclical time	Hour and month encoded with sine/cosine terms
Short memory	One-hour lag variables
Cumulative loading	Six-hour rolling means of urban temperature and radiation

2. Machine-learning framework and validation strategy

The machine-learning stage was designed as a chronological prediction problem rather than a randomly shuffled regression task. A Decision Tree served as a low-complexity, interpretable baseline, while Random Forest was used as the main model because it can represent nonlinear relationships and interactions among meteorological and temporal predictors.

Jan-Jun 2024 Proxy specification	Jul-Dec 2024 Model training	5-fold TSCV Hyperparameter search	Jan-Aug 2025 Independent test	Forecasting 15 min to 3 h
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Time-aware model development

- Training and hyperparameter optimization used July-December 2024; January-August 2025 was kept as an independent out-of-sample test period.
- Hyperparameter optimization used five sequential Time Series Cross-Validation folds to preserve temporal order and reduce information leakage.
- RandomizedSearchCV sampled 25 parameter combinations, including ensemble size, tree depth, split/leaf constraints, and feature subsampling.
- Models were developed separately for each urban station. Radiation-inclusive models were the main configuration; a no-radiation variant was retained as a sensitivity check.

Evaluation metrics and forecasting baseline

Metric	Purpose
R ²	Explained variance / overall fit
RMSE	Error magnitude with stronger penalty for large errors
MAE	Average absolute prediction error
Bias	Signed mean error to detect systematic over- or underprediction

For future prediction, horizon-specific targets were created by shifting ΔT and ΔPET_{proxy} forward in time. A persistence model - carrying the most recent observed value forward - was used as the reference benchmark. ΔPET_{proxy} was forecast at 15, 30, 45, and 60 minutes during summer; ΔT was additionally tested up to three hours ahead.

Model interpretation used permutation importance on the independent test set. Each predictor was shuffled repeatedly and the decrease in test-set R^2 was used as the measure of feature relevance.

3. Independent test-set performance

Random Forest reproduced the observed $\Delta\text{PET}_{\text{proxy}}$ variability substantially better than the single Decision Tree baseline. The independent 2025 test data show a clear linear relationship between observed and predicted values across all three urban stations, while larger errors remain at the extremes of the target range.

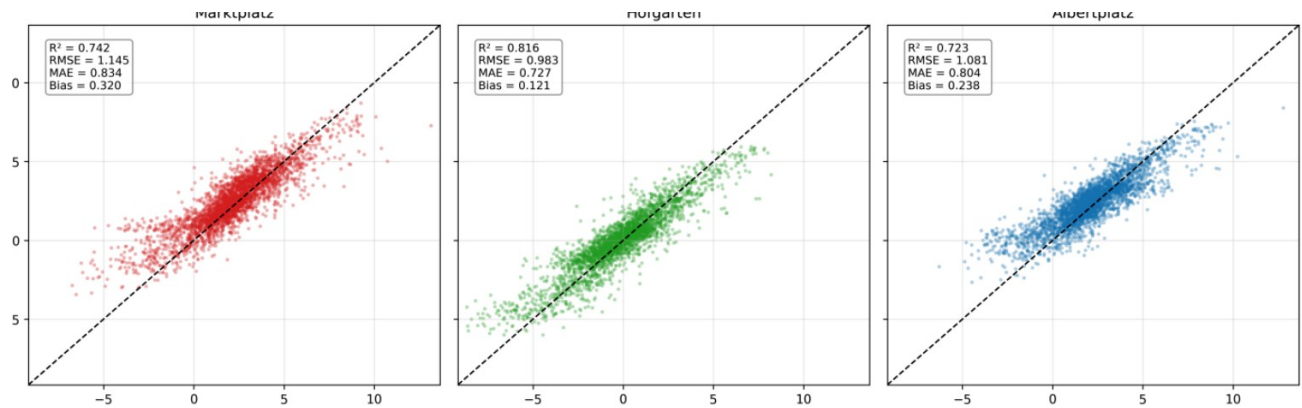


Figure 2. Random Forest observed-versus-predicted $\Delta\text{PET}_{\text{proxy}}$ on the independent 2025 test period (original thesis Figure 5.15).

Station	R^2	RMSE	MAE	Bias
Marktplatz	0.742	1.145	0.834	0.320
Hofgarten	0.816	0.983	0.727	0.121
Albertplatz	0.723	1.081	0.804	0.238

Hofgarten showed the strongest agreement, while Marktplatz and Albertplatz also maintained good predictive performance. Positive bias at all three stations indicates slight overprediction on average. The remaining scatter near the lower and upper ends of the range suggests that more extreme thermal-stress contrasts are harder to predict than central conditions.

The thesis also compared Random Forest against the Decision Tree baseline and found the ensemble model consistently more accurate, with higher R^2 and lower error metrics across the three stations. This supports the use of the ensemble approach for a dataset in which thermal contrast depends on multiple interacting meteorological drivers.

4. Short-term forecasting against persistence

The forecasting experiments tested whether the models added information beyond simply carrying the latest observation forward. This is important for high-frequency environmental time series, where persistence can be a strong baseline at very short horizons.

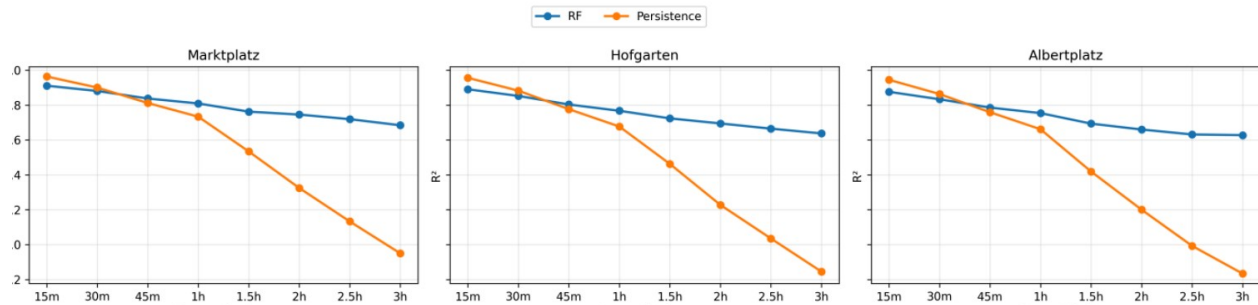


Figure 3. ΔT forecast skill up to three hours. Random Forest (blue) is compared with persistence (orange).

For ΔT , persistence is competitive at 15-30 minutes, but its skill decreases rapidly after about one hour. Random Forest declines more gradually and remains positive through the three-hour horizon at all three stations. From roughly 45 minutes to one hour onward, the model generally performs better than persistence.

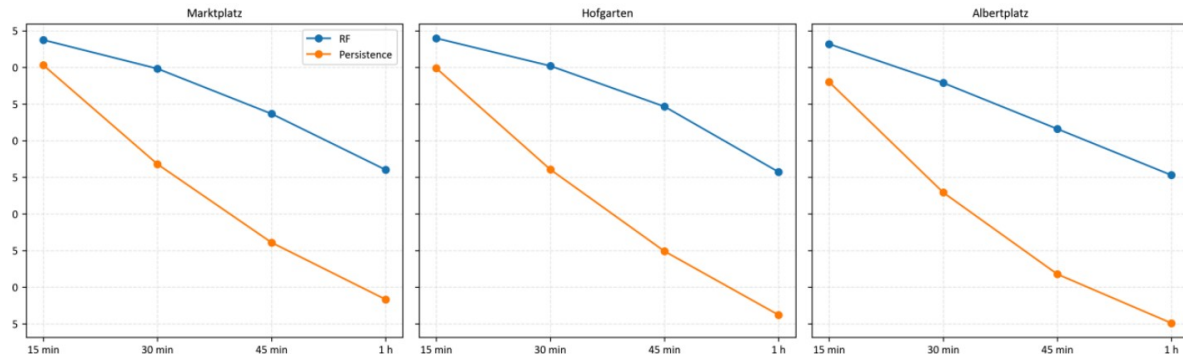


Figure 4. Summer ΔPET_{proxy} forecast skill at 15-60 minute lead times.

ΔPET_{proxy} is more difficult to forecast because it combines several meteorological components rather than a single temperature difference. Random Forest remains better than persistence, but skill weakens markedly at 45-60 minutes. The result supports using the proxy mainly for nowcasting and very-short-term heat-stress indication rather than longer-horizon prediction.

5. Model interpretation, findings, and limitations

Permutation importance was used to connect the data-driven model back to physically meaningful drivers. The ranking was broadly consistent across the three stations, while the relative importance of humidity and temperature varied with local microclimatic setting.

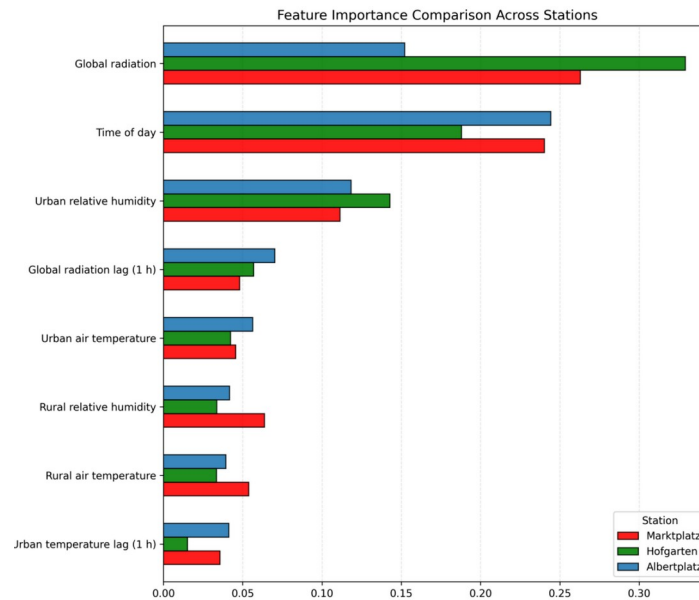


Figure 5. Permutation feature importance across the three urban stations (original thesis Figure 5.19).

Global radiation was the strongest predictor, followed by time of day. Urban relative humidity contributed moderately, while temperature variables remained relevant but less dominant. One-hour lag variables generally had lower importance, suggesting that instantaneous forcing was more influential than short-memory terms for the modeled $\Delta\text{PET}_{\text{proxy}}$ response.

Main research contributions

- Integrated a municipal urban sensor network with a standardized rural DWD reference on a common 15-minute time grid.
- Combined classical UHI intensity (ΔT) with a transparent PET-inspired thermal-stress proxy under realistic data limitations.
- Used chronological train/test separation, time-series cross-validation, independent testing, and persistence benchmarking.
- Interpreted Random Forest behavior with permutation importance and station-specific predictor rankings.

Key limitations

The PETproxy is an empirical approximation and does not replace a full physiological thermal index. Global radiation was substantially less complete than the core meteorological variables, and the fixed urban sensors represent very local surroundings that can be affected by shading, nearby walls, surface materials, and street geometry. The machine-learning models should therefore be interpreted as observation-based predictive models, not as physical urban energy-balance models.